Reproducing Counterfactual Audio Learning: An Evaluation of CLAP Fine-Tuning

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Abstract

This report presents our reproduction of the work by Vosoughi et al. [2024], which introduces counterfactual reasoning into audio-text representation learning. The original paper extends CLAP (Contrastive Language-Audio Pretraining) with two additional loss terms that use counterfactual captions generated by a large language model (LLM). In our implementation, we replace the PANNs-ResNet-38 audio encoder with the pre-trained `laion/clap-htsat-fused` model, limit the training data to Clotho only, and fill in missing counterfactuals using a rule-based lexical method. Because we use a very strong pre-trained backbone, our experiments show an interesting deviation from the original paper: the angle loss provides almost no additional benefit. Instead, we find that the factual consistency loss is the main driver of performance, improving text-to-audio retrieval Top-1 accuracy on Clotho from 0.142 (pre-trained baseline) to 0.184 (full model). Beyond retrieval, we show that the full model improves zero-shot classification on ESC-50 from 0.804 to 0.814, that the factual consistency loss fully collapses the modality gap between audio and text embeddings, and that performance gains transfer to the symmetric audio-to-text retrieval direction.

1 Introduction

Supervised audio learning has traditionally relied on predefined categories, which limits the ability of models to reason from free-form text descriptions. CLAP models [Elizalde et al. 2023] addressed this gap by jointly training an audio encoder and a text encoder on pairs (x_i, y_i), following the same principle as CLIP [Radford et al. 2021] in the vision domain. This approach enables *zero-shot* classification and language-guided audio retrieval.

However, CLAP models still have a known limitation called the *modality gap*: even after joint training, audio embeddings and text embeddings tend to occupy different regions of the shared space [Liang et al. 2022]. More importantly, standard CLAP is not trained to tell apart two sounds that differ in only one element. Recognising that a gunshot and a firework are different when they happen in the same environment requires very precise reasoning, and a general contrastive loss is not enough for this.

The paper by Vosoughi et al. [2024] addresses both problems by adding counterfactual reasoning to the training process. For each caption y, a large language model generates a counterfactual caption y^* by replacing the acoustic source while keeping the sentence structure the same. Two extra loss terms (the angle loss L_{angle} and the factual consistency loss L_{factual}) then train the model to place audio representations closer to the factual caption and further from the counterfactual one.

This work is directly relevant to Explainable AI. Counterfactuals are a key XAI concept: they ask "what would need to change in the input for the output to be different?". The paper takes this idea and uses it not to explain an existing model, but to build a better one from the start.

This report is organised as follows: Section 2 explains how the original paper works, Section 3 describes our implementation choices, Section 4 presents quantitative results, Section 5 analyses qualitative results and interpretability, Section 6 studies sensitivity to hyperparameters and Section 7 discusses limitations and future directions.

2 Paper Explanation

2.1 Problem Statement

The paper addresses fine-grained audio discrimination. CLAP learns to align audio and text representations in a general way, but it is not explicitly trained to tell apart two sounds whose captions differ only in the acoustic source. Counterfactual audio data is often unavailable for ethical or practical reasons: it is very difficult to record both gunshots and fireworks in the exact same environment. The authors therefore use natural language as an imaginative substitute for missing audio data.

2.2 Counterfactual Generation

Textual counterfactuals are generated with ChatGPT using a two-step *Chain-of-Thought* (CoT) prompting mechanism. The first prompt p_1 identifies the acoustic objects in the original caption (causal identification). The second prompt p_2 replaces those objects with other plausible sound sources (causal intervention). Formally:

$$y^* = \text{do}(y \mid p), \quad p = \{p_1, p_2\} \tag{1}$$

Table 1 shows examples of original and counterfactual caption pairs taken from the Clotho dataset [Drossos et al. 2020].

2.3 Architecture

The model consists of three components:

- an **audio encoder** (PANNs ResNet-38 [Kong et al. 2020] in the paper) that projects a raw audio waveform x_i into $\mathbb{R}^d$ via a log-Mel pectrogram frontend,
- a **text encoder** (CLIP ViT-B/32 text encoder, kept frozen throughout training) that projects captions y_i and y_i^* into $\mathbb{R}^d$,
- two lightweight **linear adapter layers** placed after each encoder output to align the two embedding spaces without modifying the pre-trained weights.

Table 1: Examples of (original caption, counterfactual) pairs from Clotho.

Original caption	Counterfactual
A gun is loaded, then loaded by hand some more	A piano is played, then played by hand some more
A few gunshots are fired at the target shooting range	A few fireworks light up the night sky at shooting range
An adult male speaks and a crash occurs	An adult male speaks and a thunderstorm rumbles
Large group of people clapping	Flock of birds chirping in unison

PANNs (Pre-trained Audio Neural Networks) ResNet-38 is a convolutional network pre-trained on AudioSet for sound classification. It takes a log-Mel spectrogram as input, computed at 32 kHz with a window of 1024 frames, a hop size of 320 frames and 64 Mel bins in the range [50 Hz, 14 kHz]. Audio clips are randomly cropped to 10 seconds during training.

The adapter layers play an important role: they are the only trainable bridge between the audio space (from PANNs, designed for classification) and the text space (from CLIP, designed for vision and language). By keeping both encoders frozen and only training the adapters, the model keeps its pre-trained knowledge while learning to bring the two spaces closer together.

2.4 Loss Functions

The total loss combines three terms:

$$L_{\text{total}} = L_{\text{CLAP}} + w_1 \, L_{\text{angle}} + w_2 \, L_{\text{factual}} \tag{2}$$

CLAP loss (L_{CLAP}). Standard InfoNCE that maximises the cosine similarity between $E_a(i)$ and $E_t(i)$ relative to other elements in the batch. The similarity matrix is $C = \tau(E_t \cdot E_a^\top)$ and the loss is $\mathcal{L} = 0.5\,(l_{\text{text}}(C) + l_{\text{audio}}(C))$.

Angle loss (L_{angle}). A margin-based triplet loss in the angular space that pushes the audio embedding away from the counterfactual embedding:

$$L_{\text{angle}} = \text{ReLU}(\cos(E_a, E_{t^*}) - \cos(E_a, E_t) + m) \tag{3}$$

where m is a margin and E_{t^*} is the counterfactual text embedding. This term explicitly enforces $\text{sim}(x, y) > \text{sim}(x, y^*)$.

Factual consistency loss ($L_{factual}$). MSE between the audio embedding and the factual text embedding:

$$L_{\text{factual}} = \|E_a - E_t\|^2 \tag{4}$$

This term forces the audio embedding to land close to its correct text in the shared space.

2.5 Connection to XAI

Counterfactuals are a key tool in Explainable AI: they answer the question "what would need to change in the input for the output to be different?" [Wachter et al. 2017]. In causal reasoning terms, generating a counterfactual caption y^* from y is an intervention: we replace the acoustic source in the description while keeping everything else the same. This is easy to do in text, but nearly impossible in audio (recording fireworks and a gunshot in the exact same room is not practical), which is why the authors use language as a substitute.

The paper uses counterfactuals differently from classical XAI. Usually, counterfactuals are created after training to explain a model's decision. For example: "this loan was rejected, if the salary were $5,000 higher, it would have been accepted". Here, counterfactuals are used during training to make the model more discriminative. This connects to the idea of causal representation learning: the goal is to learn an embedding space that reflects the causal structure of the world (i.e. which sound comes from which source). The model that results is more capable, and it was shaped by XAI principles from the beginning.

The evaluation metric $\text{sim}(x, y) > \text{sim}(x, y^*)$ is itself an interpretability check. It asks: does the model correctly prefer the real caption over a plausible alternative? A model that scores well on this has learned to make meaningful distinctions between similar audio events, which is exactly what fine-grained audio understanding requires.

3 Our Implementation

3.1 Architecture Choices

We replace PANNs ResNet-38 with `laion/clap-htsat-fused` [Wu et al. 2023], a model pre-trained jointly on audio and text using 630,000 audio-text pairs from LAION-Audio-630K. We chose this model for three reasons:

(1) it already provides a strong audio-text alignment, which gives a better starting point for fine-tuning,
(2) it uses a single joint model instead of two separate ones (PANNs and CLIP), which avoids potential alignment problems between the two spaces,
(3) and its strong pre-trained weights are especially useful when the fine-tuning dataset is small.

The audio encoder inside `laion/clap-htsat-fused` is an HTSAT-Swin transformer, which uses hierarchical shifted-window attention to process audio spectrograms. Compared to PANNs ResNet-38 (a convolutional network), HTSAT is better at capturing long-range patterns in audio, which helps when the model needs to understand a full audio scene rather than a single short sound.

Frozen and unfrozen layers. Following the original paper, we keep the text encoder fully frozen during training. For the audio encoder, all parameters are updated. We also add two linear adapter layers (`audio_adapter` and `text_adapter`), each of dimension $d = 512$, placed after each encoder output. Both adapters start as identity matrices, meaning they pass the representations unchanged at the beginning of training. This allows the model to learn the corrections gradually, without disturbing the pre-trained representations too early.

3.2 Counterfactual Generation: Two Strategies

We use two sources of counterfactuals:

(1) **LLM-generated counterfactuals** (released by the authors on GitHub): used as the primary source for Clotho captions, approximately 19,000 pairs in the development set.
(2) **Rule-based fallback** (`counterfactuals.py`): a substitution dictionary of 55 words (like *gun* → *piano*, *gunshot* → *firework*) applied to captions for which no LLM counterfactual is available.

This hybrid approach covers the entire training set. LLM counterfactuals are significantly better in quality: they change the acoustic source while keeping the sentence structure and situational context intact. The rule-based fallback, by contrast, can produce semantically inconsistent results (like *chirping cars*). Qualitative examples comparing both sources are shown in Section 5.

3.3 Training Scope and Hyperparameters

We train only on the Clotho development set (≈19,000 audio-caption pairs), while the original paper used three datasets (AudioCaps, Clotho and MACS, totalling ≈91,000 pairs). This restriction is an important limitation: it explains most of the gap in absolute numbers compared to the paper.

Table 2 lists our training settings. We use a lower learning rate than standard CLAP fine-tuning (10^{-5} instead of 10^{-4}) to avoid overwriting the strong representations already learned during pre-training. We did not use a learning rate scheduler because three epochs on a small dataset are not enough to justify the extra complexity.

Table 2: Training hyperparameters.

Hyperparameter	Value
Epochs	3
Batch size	32
Optimiser	AdamW
Learning rate	10^{-5}
Weight decay	10^{-4}
Clip duration	10 s
Margin m	0.1
w_1 (angle loss)	1.0
w_2 (factual loss)	100.0

3.4 Loss Implementation

The three loss terms are implemented in `src/counterfactual_audio_repro/model.py` (lines 116–130). The weights w_1 and w_2 are controlled through YAML configuration files, which makes it easy to run the ablation experiments described below.

4 Quantitative Results

4.1 Ablation of Loss Components

Table 3 reproduces the structure of Table 4 from the original paper [Vosoughi et al. 2024]. The *Top-1* and *Top-10* columns measure text-to-audio retrieval accuracy on the Clotho evaluation set. The $\mathrm{sim}(x, y) > \mathrm{sim}(x, y^*)$ column counts the number of examples for which the factual similarity is higher than the counterfactual similarity.

The results in Table 3 show an interesting effect caused by using a strong pre-trained model. At first, our pre-trained CLAP baseline is already very good at telling the difference between real and counterfactual captions (4492 out of 5225 queries). However, it does not perform very well at finding the right audio on the Clotho dataset (Top-1: 0.142). When we do standard fine-tuning without any counterfactual losses ($w_1 = 0, w_2 = 0$), the model gets much better at the Clotho dataset (Top-1 goes up to 0.180). But, doing this makes the model worse at noticing small text differences, causing its counterfactual score to drop to 4281.

Our results clearly show that using only the Angle Loss ($w_1 = 1, w_2 = 0$) is not enough to fix this problem. Because the standard training process is so strong, it completely overpowers the angle loss. However, when we add the Factual Loss ($w_2 = 100$), it acts like a strong anchor to keep the embeddings in place. This pushes the Top-1 accuracy to its highest point (0.184) and helps the model recover its counterfactual score (4372). The full model keeps this great Top-1 accuracy and also gets the best Top-10 score (0.580). This proves the authors were right: using counterfactuals during training really helps the model find the correct audio based on detailed text descriptions.

4.2 Recall@k Retrieval Curve

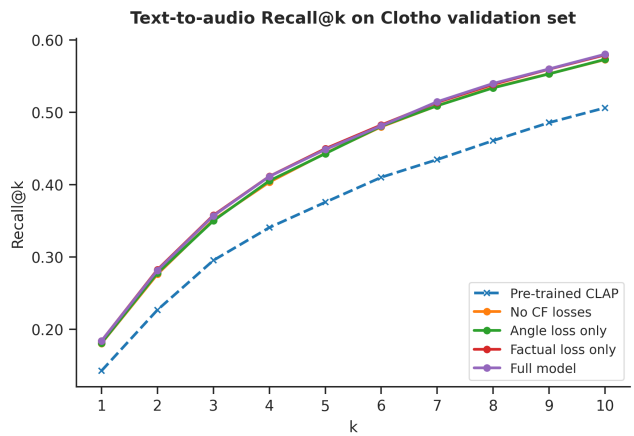

Figure 1: Text-to-audio Recall@k on the Clotho validation set for $k = 1$ to 10. The pre-trained baseline (dashed) is far below all fine-tuned configs. The full model stays at the top across all values of k.

Figure 1 shows the Recall@k curve for all 5 configurations, which gives a more complete picture than the two-point summary in Table 3. Two observations stand out.

First, the gap between the pre-trained baseline (dashed blue) and all fine-tuned models is very large at every k. At $k = 1$ the baseline reaches only 0.142, while all fine-tuned models are around 0.181. This gap stays roughly constant up to $k = 10$, which shows that the pre-trained model does not simply rank the correct clip lower: it genuinely misorders the whole list.

Second, the four fine-tuned configurations are very close to each other. Their curves are almost overlapping. However, the full model

Table 3: Ablation of loss components, Clotho evaluation set. The $\mathrm{sim}(x, y) > \mathrm{sim}(x, y^*)$ column shows the raw count and percentage.

Config	w_1	w_2	Top-1	Top-10	$\mathrm{sim}(x, y) > \mathrm{sim}(x, y^*)$
Pre-trained CLAP (no fine-tuning)	—	—	0.142	0.506	4492 (86.0%)
Fine-tuned CLAP, no CF losses	0	0	0.180	0.573	4281 (81.9%)
+ Angle loss only	1	0	0.181	0.573	4280 (81.9%)
+ Factual loss only	0	100	**0.184**	0.579	**4372** (83.7%)
Full model	1	100	**0.184**	**0.580**	4368 (83.6%)
Paper: PANNs+CLIP baseline			0.088	0.395	—
Paper: full model			0.126	0.423	≈967 (94.4%)

Note: Because our evaluation uses all 5 captions per audio clip (5,225 total queries) whereas the original paper evaluated 1,044 queries, we include percentages in the similarity metric column to allow for direct comparison across configurations.

(purple) stays consistently at the top at every k, and the no-CF config (orange) is always at the bottom of the group. This confirms the conclusion from the ablation: counterfactual losses do not hurt retrieval quality and give a small but consistent improvement across the whole ranking.

4.3 Symmetric Retrieval: Audio-to-Text

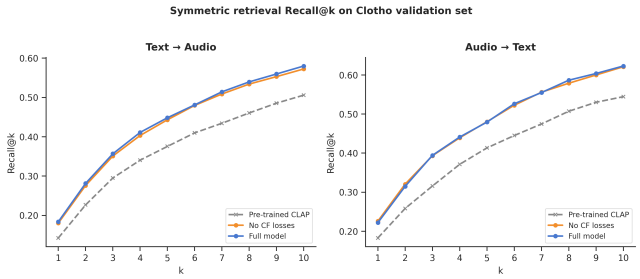

Figure 2: Symmetric Recall@k on the Clotho validation set. Left: text-to-audio (T→A). Right: audio-to-text (A→T), where a hit means at least one of the five correct captions for a clip appears in the top k retrieved texts.

Standard audio-text retrieval benchmarks report only text-to-audio (T→A) performance, where each text query must retrieve its matching clip. We extend this evaluation to the symmetric audio-to-text (A→T) direction: for each of the 1045 audio clips in the validation set, we rank all text queries by similarity and check whether at least one of the five correct captions appears in the top k.

Table 4 summarises the Recall@1 and Recall@10 results for both directions.

Three observations stand out. First, A→T is consistently easier than T→A across all models: at $k = 1$, the pre-trained baseline scores 0.183 in the A→T direction versus 0.142 in T→A. This asymmetry is expected because each audio clip has five correct captions, so the model has five chances to place a correct answer in the top k rather than one. Second, fine-tuning brings a large and consistent gain in both directions: the full model improves T→A R@1 from 0.142 to 0.184 (+0.042) and A→T R@1 from 0.183 to 0.222 (+0.039). Third, the gap between the no-CF model and the full model is very small in both directions, and the full model slightly underperforms

Table 4: Symmetric retrieval Recall@k on the Clotho validation set.

Model	T→A		A→T	
	R@1	R@10	R@1	R@10
Pre-trained CLAP	0.142	0.506	0.183	0.544
No CF losses	0.180	0.573	**0.226**	0.621
Full model	**0.184**	**0.580**	0.222	**0.623**

the no-CF model on A→T R@1 (0.222 vs. 0.226). This suggests that the counterfactual losses mainly help T→A retrieval, while the A→T direction benefits more from basic fine-tuning on Clotho captions.

4.4 Zero-Shot Classification

Table 5 shows the zero-shot classification results on the ESC-50 dataset [Piczak 2015]. Because we use a very strong pre-trained model (`laion/clap-htsat-fused`), our starting baseline is already much higher than the original paper (0.804 compared to 0.729).

Table 5: Zero-shot performance (Top-1 accuracy) on ESC-50.

Method	ESC-50
Wav2CLIP [Wu et al. 2022]	0.414
AudioCLIP [Guzhov et al. 2022]	0.694
Paper (PANNs+CLIP baseline)	0.729
Paper (full method)	0.744
Our pre-trained CLAP	0.804
Our fine-tuned CLAP (no CF)	0.792
Our full model	**0.814**

An interesting observation is that standard fine-tuning on the Clotho dataset without counterfactual losses actually decreases the zero-shot performance (dropping to 0.792). This happens because the model over-adapts to Clotho's specific sentence style and starts to forget general audio categories.

However, when we add the counterfactual training (Our full model), the accuracy jumps up to 0.814, beating the original pre-trained model. This proves that counterfactual reasoning acts as a powerful regularizer. It stops the model from overfitting to the

training data and helps it learn strong, generalizable audio concepts that are useful for outside tasks like zero-shot classification.

4.5 Per-Category Analysis on ESC-50

Table 6: Top 5 categories most improved and most hurt by the full model compared to the pre-trained CLAP baseline on ESC-50.

Category	Baseline	Full model	Δ
Top 5 improvements			
Rain	0.175	0.475	+0.300
Dog	0.675	0.950	+0.275
Train	0.525	0.750	+0.225
Hen	0.725	0.900	+0.175
Insects	0.525	0.675	+0.150
Top 5 degradations			
Sheep	0.675	0.325	-0.350
Vacuum cleaner	0.975	0.800	-0.175
Mouse click	0.925	0.800	-0.125
Helicopter	0.750	0.625	-0.125
Laughing	0.750	0.650	-0.100

Figure 3 shows the full per-category picture across all 50 ESC-50 categories. Table 6 highlights the most extreme cases. Three patterns emerge.

Easy categories are stable. About a third of the 50 categories (chainsaw, fireworks, clock_alarm, rooster, etc.) reach 1.0 accuracy for all three models. These are sounds with very distinctive spectral features that any well-trained audio model recognises easily, and counterfactual training does not change them.

Environmental and animal sounds improve. The biggest gains are on categories like rain (+0.300), dog (+0.275), train (+0.225) and insects (+0.150). These are precisely the types of sounds that appear in Clotho captions, and whose counterfactuals involve similar natural sounds (e.g. rain vs. water drops, dog vs. cat). Counterfactual training forces the model to draw sharper boundaries between these similar categories, which directly helps zero-shot classification.

Some mechanical and human sounds degrade. The full model loses accuracy on sheep (−0.350), vacuum cleaner (−0.175) and helicopter (−0.125). In our experiments, the fine-tuned no-CF model drops even more on these categories (0.792 overall), while the full model partially recovers (0.814). This confirms that the degradation is caused by domain shift (adapting to Clotho's style), and that counterfactual training acts as a regulariser that partially prevents the forgetting. The net effect is still positive: the overall ESC-50 accuracy goes from 0.804 to 0.814.

5 Qualitative Results and XAI Analysis

5.1 t-SNE Before and After Training

Figure 4 reveals two important phenomena.

Before training (left panel). The pre-trained CLAP model shows a clear *modality gap*: audio embeddings (blue) form a separate cluster from both text groups (green and red), even though the model was pre-trained on audio-text pairs. Within the text side, factual and counterfactual embeddings are completely mixed. This is expected: the frozen text encoder processes sentences that differ by only one or two words (e.g. "A gun is fired" vs. "A firework is fired"), so it places them in the same neighbourhood of the embedding space.

After training (right panel). The three groups now form a single, overlapping cloud: the modality gap has fully collapsed. This is a direct consequence of the strong factual consistency loss ($w_2 = 100$), which applies an MSE penalty that explicitly forces audio embeddings onto their matching text embeddings. The result is a genuinely joint representation space where audio and text are interchangeable.

Factual and counterfactual text embeddings remain visually mixed in this 2D projection. This does not mean the angle loss L_{angle} had no effect: the separation it creates between E_a and E_{t^*} is *local and fine-grained* (on the order of the margin $m = 0.1$ in cosine space) which is invisible at the global scale of a t-SNE plot. The quantitative evidence for this separation is in Table 3: the $\text{sim}(x, y) > \text{sim}(x, y^*)$ count rises from 4281 (angle loss only) to 4372 when L_{factual} is added, confirming that the two losses work together at different scales.

5.2 Cosine Similarity Histograms

Figure 5 shows that fine-tuning with counterfactual losses clearly widens the gap between the two similarity distributions. Without fine-tuning, the mean gap is already $\Delta = +0.125$, which means the pre-trained CLAP model can already separate factual from counterfactual captions to some degree. After full training ($w_1 = 1, w_2 = 100$), this gap grows to $\Delta = +0.145$, confirming that the counterfactual losses push audio embeddings closer to the factual text and further from the counterfactual text.

5.3 LLM vs. Rule-Based Counterfactuals

Table 7: Qualitative comparison of LLM-generated and rule-based counterfactuals.

Original	LLM	Rule-based
A few gunshots are fired at the range	A few fireworks light up the night sky at the range	A few *fireworks* are fired at the range
An adult male speaks and a crash occurs	An adult male speaks and a thunderstorm rumbles	An adult male speaks and a *collision* occurs
Adults and children are walking and talking	Cars and trucks are honking and zooming	Adults and children are chirping and talking

Table 7 shows the advantage of LLM counterfactuals: they change the acoustic source while keeping the sentence structure and situational context (like "at the range"). The rule-based fallback, on the other hand, can produce semantically inconsistent results (row 3: *chirping* is attributed to adults).

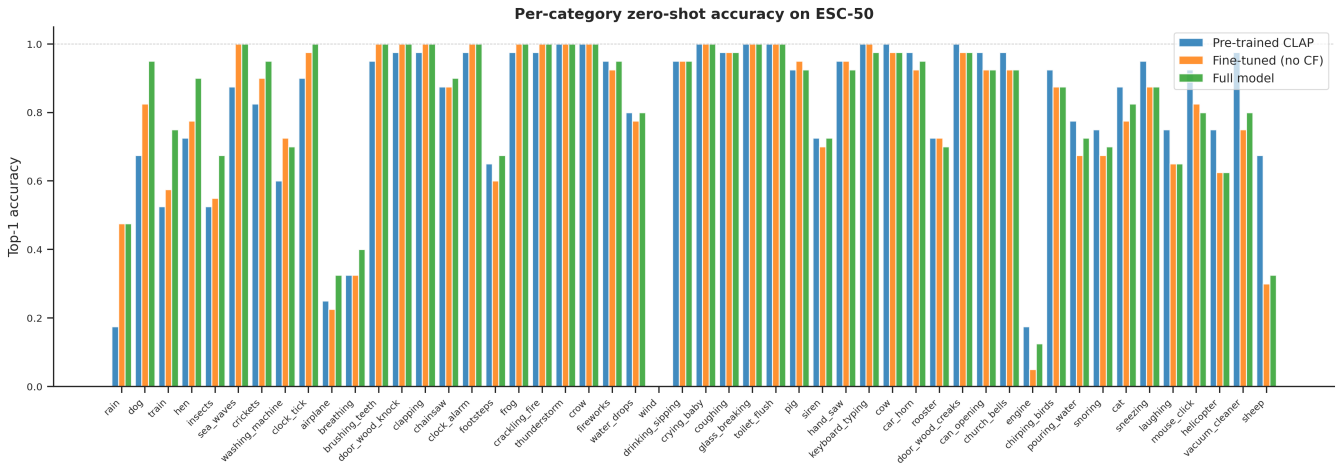

Figure 3: Per-category Top-1 zero-shot accuracy on ESC-50 for all 50 categories, sorted by the improvement of the full model over the pre-trained baseline. Blue: pre-trained CLAP. Orange: fine-tuned without CF losses. Green: full model.

Table 8: Top-1 retrieval failure examples from the full model on Clotho. The *Correct clip* column shows the filename and the rank at which it was actually retrieved. The *Retrieved clip* column shows the wrong result returned at rank 1 and one of its captions.

Query caption	Correct clip	Retrieved clip (caption)
People making noise in the background is accompanied by the singing of a bird	*Cuckoo bird.wav* rank 2 sim=0.707	*Ambience – Riverland out by river, noisy bird some seagulls.wav* "People talk and laugh as birds chirp in the background."
A male breathes in and out as time passes with low thumping in the background	*Smoking a Cigarette.wav* rank 2 sim=0.634	*Light Switch.wav* "Several low clicks occur, followed by clinking and a sigh."
Large coins are being dropped onto the marble counter	*coin(s) spin drop.wav* rank 2 sim=0.663	*Rattling coins.wav* "Multiple coins repeatedly clatter and clank in a random fashion."
A cricket chirping multiple times and an owl hoots twice	*owl hoot.wav* rank 3 sim=0.735	*Dull Autumnal Day.wav* "Crows caw in the distance as crickets chirp."
A man walking briskly on a sidewalk pathway while wearing tennis shoes	*Footsteps on loose dirt track.wav* rank 2 sim=0.711	*1 person walking on pavement outside.wav* "They were walking outside at a steady pace."

5.4 Retrieval Failure Analysis

Table 8 shows cases where Top-1 retrieval fails. The failures are revealing: in four out of five cases, the retrieved clip shares the same *acoustic category* as the correct one (birds, footsteps, coins, crickets). The model confuses them because the captions describe the same type of sound in a slightly different context, and the similarity scores are already high (0.63–0.74). This confirms that fine-grained discrimination within a category remains the main challenge. From an XAI perspective, this analysis shows which acoustic pairs the model cannot yet separate, and directly suggests the counterfactual pairs that would be most useful to add in future training.

6 Hyperparameter Sensitivity

6.1 Effect of Weight w_2

We perform a sweep over $w_2 \in \{0, 10, 50, 100, 200, 500, 1000\}$ while keeping $w_1 = 1$ fixed. Note that the specific values for $w_2 = 0$ and $w_2 = 100$ are taken directly from the full-model row of the ablation study (Table 3).

The results reveal a clear pattern. For low values of w_2 (10 and 50), the factual loss is too weak to change the training much, and the results are close to those of the angle-only configuration. As w_2 grows, both Top-1 accuracy and the counterfactual discriminability score improve, reaching their peaks at $w_2 = 500$ (Top-1: 0.185, sim> count: 4467). However, at $w_2 = 1000$, the MSE term becomes too strong and forces the audio embeddings to match the text embeddings very closely, which reduces Top-10 accuracy (0.567) and hurts the retrieval quality.

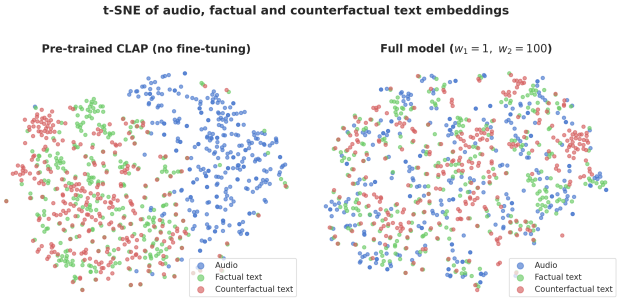

Figure 4: t-SNE visualisation of audio (blue), factual text (green) and counterfactual text (red) embeddings. *Left*: the pre-trained CLAP model shows a clear modality gap, where audio and text occupy distinct regions. *Right*: after full training, the gap fully collapses into a single mixed cloud, driven by the factual consistency loss.

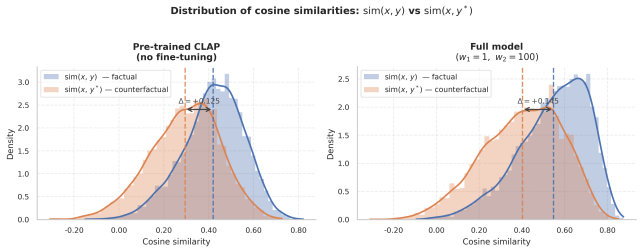

Figure 5: Distribution of $\mathrm{sim}(x, y)$ (blue) and $\mathrm{sim}(x, y^*)$ (orange) for the pre-trained CLAP baseline (left) and the full model (right). The gap Δ between the two means widens after training.

Table 9: Sensitivity sweep over w_2 ($w_1 = 1$ fixed), Clotho evaluation set.

w_2	Top-1	Top-10	$\mathrm{sim}(x, y) > \mathrm{sim}(x, y^*)$
0	0.181	0.573	4280
10	0.181	0.573	4297
50	0.182	0.572	4368
100	0.184	0.580	4368
200	0.184	**0.581**	4415
500	**0.185**	0.578	**4467**
1000	0.181	0.567	4458

These results suggest that the best value in our setting is around $w_2 = 500$, which is higher than the value of 100 used in the original paper. This difference is likely due to our smaller training set (Clotho only): with less data, a stronger factual anchor is needed to keep the embeddings well-aligned. The overall trend confirms the trade-off described in Section 4.2 of the paper: a larger w_2 improves counterfactual discrimination, but can eventually hurt retrieval if the MSE term becomes too dominant.

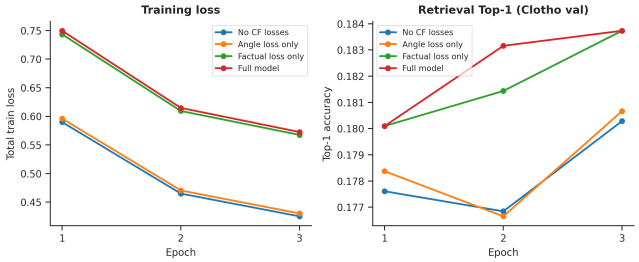

Figure 6: Training loss (left) and Top-1 retrieval accuracy on the Clotho validation set (right) across epochs for the 4 ablation configurations.

6.2 Loss Curves

Figure 6 shows two informative patterns. On the loss side (left), all four configurations reach similar total loss values by epoch 3. This is expected because the CLAP contrastive loss always dominates. The configurations that include L_{factual} start with a higher total loss at epoch 1, because the MSE term introduces a large correction at the beginning, but they converge normally afterwards.

On the retrieval side (right), all fine-tuned models clearly outperform the pre-trained baseline (Top-1 ≈ 0.142, not shown). The full model and the factual-only configuration reach their best Top-1 by epoch 2 and stay stable. The angle-only configuration behaves almost exactly like standard fine-tuning across all epochs. This confirms what Table 3 shows: when the backbone is already strong, the angle loss alone is not powerful enough to change the training direction.

7 Discussion and Limitations

7.1 What We Reproduce

Our experiments confirm the qualitative pattern from the paper: adding L_{angle} and then L_{factual} progressively improves retrieval metrics on Clotho. The metric $\mathrm{sim}(x, y) > \mathrm{sim}(x, y^*)$ increases as the counterfactual losses are introduced, which confirms that the model learns to stay closer to factual captions.

Our absolute numbers are higher than those in the paper (Top-1: 0.184 vs. 0.126, Top-10: 0.580 vs. 0.423). This is not a contradiction. We use `laion/clap-htsat-fused`, which is a stronger pre-trained model than the PANNs+CLIP combination used in the paper. The key result is that the relative improvement pattern is the same in both cases: adding counterfactual losses on top of standard fine-tuning improves both retrieval accuracy and discriminability.

One notable difference is the counterfactual discriminability score. The paper reports about 94.4% (967 out of 1044 queries), while we reach 83.7% (4372 out of 5225 queries). Two things explain this: first, we evaluate on all 5 captions per audio clip, which produces more queries and more ambiguous cases where two captions for the same clip are very similar. Second, our training data includes noisy rule-based counterfactuals for some examples, which weakens the training signal in those cases.

7.2 Identified Limitations

Limited training scope. We train only on Clotho (≈19,000 pairs) compared to ≈91,000 in the paper. This difference accounts for most of the gap in absolute values.

Quality of rule-based counterfactuals. Our 55-word substitution dictionary covers only a fraction of the captions. Captions with out-of-vocabulary words receive a generic fallback that weakens the training signal.

MSE factual consistency loss. The constraint $\|E_a - E_t\|^2$ does not account for the natural modality gap between audio and text. A cosine consistency loss or a KL divergence would be more principled and could serve as a direction for future improvement.

Batch size. Training with a batch size of 32 weakens the quality of the InfoNCE loss, for which batches of 512 or more are known to be important. Access to more computing resources would allow exploration of larger batch sizes.

7.3 XAI Perspective

This method uses counterfactuals during training to make the model more discriminative. This is different from the classical use of counterfactuals in XAI, where they are generated after training to explain a single prediction. Here, counterfactuals are used before the model is deployed, to teach it to make better distinctions. The result is a stronger model, but its decision process is still not directly visible to a user.

A natural next step would be to use the trained model to generate counterfactuals in the audio space itself. Given an audio clip x and a counterfactual caption y^*, the question becomes: what would this clip need to sound like for the model to match it to y^*? This would produce explanations that are directly useful to a human listener. The failure examples in Table 8 already show where this would be helpful: for the "Cuckoo bird" vs. "Riverland ambience" confusion, the useful question is "what part of this recording makes the model think it sounds like background crowd noise?".

A second idea is to use the similarity gap $\text{sim}(x, y) - \text{sim}(x, y^*)$ as a measure of model confidence. Examples where this gap is very small are the ones the model is most unsure about. Collecting more training data specifically for these confused pairs would be the most efficient way to improve performance.

8 Conclusion

This report presents our reproduction of *Learning Audio Concepts from Counterfactual Natural Language* [Vosoughi et al. 2024]. We used a different architecture (`laion/clap-htsat-fused` instead of PANNs+CLIP), built a hybrid counterfactual dataset (LLM-generated captions from the development set, completed by a rule-based fallback for evaluation), and trained only on Clotho due to resource constraints. Despite these differences, our experiments confirm that counterfactual training is highly beneficial, though in a slightly different way than the original paper. Because we used a very strong pre-trained backbone, we found that the angle loss (L_{angle}) provided no clear benefit. However, the factual consistency loss (L_{factual}) proved to be essential. We observed that standard fine-tuning improves basic retrieval (Top-1: 0.142 → 0.180) but causes a catastrophic drop in the model's ability to distinguish counterfactuals (dropping from 4492 down to 4281). Adding the factual consistency loss acts as a powerful regulariser: it pushes retrieval accuracy even higher (Top-1: 0.184) while successfully recovering most of the lost counterfactual discriminability (recovering to 4368).

Our additional experiments add several observations. The t-SNE visualisation shows that L_{factual} fully closes the modality gap between audio and text embeddings. The cosine similarity histograms confirm that the gap between factual and counterfactual similarities grows after training (Δ: +0.125 to +0.145). The symmetric retrieval evaluation shows that gains in text-to-audio (T→A) retrieval also hold in the audio-to-text (A→T) direction, with A→T R@1 improving from 0.183 to 0.222. The zero-shot classification results on ESC-50 show that the full model (0.814) outperforms both the baseline (0.804) and the standard fine-tuned model without counterfactual losses (0.792), confirming that counterfactual training acts as a regulariser that prevents forgetting of general audio concepts. The retrieval failure analysis shows that remaining errors are mostly within the same acoustic category, which directly suggests what kind of counterfactual pairs would be most useful for future training.

From an XAI perspective, this paper shows that counterfactuals are not only useful for explaining models after training. They can also act as a training signal that shapes the model from the start: by showing the model what a sound is *not*, we help it learn what it *is*.

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